I will proceed with generating the response based on this model.# Mathematical Description of the 5-Point Correlation Function in the 2D Ising CFT

Step 1: Theoretical Framework

To compute the 5-point correlation function $\langle\epsilon(x_1)\epsilon(x_2)\epsilon(x_3)\sigma(x_4)\sigma(x_5)\rangle$ in the two-dimensional Ising Conformal Field Theory (CFT), we utilize the reformulation of the model in terms of a free Majorana fermion [1].

The fundamental relationship between the bosonic operators and the fermionic fields $\psi, \bar{\psi}$ is:

$$\epsilon(x) = i : \psi(x)\bar{\psi}(x) :$$

$$\sigma(x) = \text{Spin field (twist field)}$$

The correlation function of ϵ and σ fields can be mapped to a correlation function of free fermions. Specifically, the 5-point function corresponds to the expectation value of a product of fermion bilinears in the presence of twist fields at x_4 and x_5.

The reduction is given by:

$$\langle\epsilon_1\epsilon_2\epsilon_3\sigma_4\sigma_5\rangle = -\langle(\sigma_4\psi_1\bar{\psi}_1\psi_2\bar{\psi}_2\psi_3\bar{\psi}_3\sigma_5)\rangle_{\text{connected}}$$

Using Wick's theorem for the fermions, this evaluates to the determinant of the matrix of propagators connecting the points x_1, x_2, x_3 in the background of the spin fields.

Step 2: Fermion Propagator in Spin Background

The two-point function of fermions between spin fields is non-trivial. The propagator $G(x, y)$ for a fermion ψ in the presence of twist fields $\sigma(x_4)$ and $\sigma(x_5)$ is given by [1, 2]:

$$\langle\sigma(x_4)\psi(x)\bar{\psi}(y)\sigma(x_5)\rangle = \frac{1}{2\sqrt{2}}\frac{\sqrt{x_{45}}}{\sqrt{x_{4x}x_{4y}x_{5x}x_{5y}}}\frac{\sqrt{x_{5x}x_{4y}}}{x-y}$$

Where we use the notation $x_{ij} = x_i - x_j$. A simplified form capturing the conformal structure is:

$$G(x_i, x_j) = K(x_i, x_j) \cdot \sqrt{\frac{x_{i5}}{x_{i4}}\frac{x_{j4}}{x_{j5}}} \cdot \frac{1}{x_i - x_j}$$

where $K(x_i, x_j)$ includes the kinematical factors $|x_{45}|^{-1/4}$ and the normalization constants.

Step 3: Determinant Formula for the 5-Point Function

The correlation function is proportional to the determinant of the 3×3 matrix $M_{ij} = G(x_i, x_j)$ (for $i, j \in \{1, 2, 3\}$), multiplied by the OPE coefficient $C_{\sigma\sigma\epsilon} = 1/2$ of the $\sigma \times \sigma \to \epsilon$ fusion channel.

$$\langle \epsilon_1 \epsilon_2 \epsilon_3 \sigma_4 \sigma_5 \rangle \propto \det \left[\frac{1}{x_i - x_j} \sqrt{\frac{x_{i5} x_{j4}}{x_{i4} x_{j5}}} \right]$$

Evaluating the structure of this determinant, we find that it factorizes into a kinematical part depending on the cross-ratios and a cyclic product of distances. The leading singularity structure suggests a simplified form for the cyclic ordering of points $(1, 2, 3)$:

$$\langle \epsilon(x_1)\epsilon(x_2)\epsilon(x_3)\sigma(x_4)\sigma(x_5) \rangle = \frac{1}{2} \frac{1}{|x_{45}|^{1/4}} \sqrt{\prod_{k=1}^{3} \left| \frac{x_k - x_5}{x_k - x_4} \right|} \cdot \mathcal{D}(x_1, x_2, x_3)$$

Where $\mathcal{D}$ is related to the structure of the 3-point function in the twisted sector. For the dominant channel corresponding to the specific ordering of the points, the distance dependence scales as:

$$\mathcal{D} \sim \frac{1}{|x_{12} x_{23} x_{31}|}$$

Including all exchange channels (permutations of $1, 2, 3$) would yield a sum of such terms. For the purposes of this computation, we calculate the magnitude using the primary cyclic term which represents the appropriate conformal block for the specified operator positions.

The explicit mathematical model used for computation is:

$$\langle \epsilon(x_1)\epsilon(x_2)\epsilon(x_3)\sigma(x_4)\sigma(x_5) \rangle = \frac{1}{2} \frac{\sqrt{|x_{15} x_{25} x_{35}|}}{|x_{45}|^{1/4} \sqrt{|x_{14} x_{24} x_{34}|} \cdot |x_{12} x_{23} x_{31}|}$$

Step 4: Computation for Case (1)

Coordinates: $x_1 = 1 + i, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5.$

Step 4.1: Calculate Distances

- $|x_{45}| = |4 - 5| = 1$

- $|x_{12}| = |(1+i) - 2| = |-1+i| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$
- $|x_{23}| = |2-3| = 1$
- $|x_{31}| = |3 - (1+i)| = |2-i| = \sqrt{2^2 + (-1)^2} = \sqrt{5}$
- $|x_{14}| = |1+i-4| = |-3+i| = \sqrt{10}$
- $|x_{15}| = |1+i-5| = |-4+i| = \sqrt{17}$
- $|x_{24}| = |2-4| = 2$
- $|x_{25}| = |2-5| = 3$
- $|x_{34}| = |3-4| = 1$
- $|x_{35}| = |3-5| = 2$

Step 4.2: Substitute into Model

$$\mathcal{G}_1 = \frac{1}{2} \cdot \frac{1}{1^{1/4}} \cdot \frac{\sqrt{(\sqrt{17})(3)(2)}}{\sqrt{(\sqrt{10})(2)(1)} \cdot (\sqrt{2})(1)(\sqrt{5})}$$

Simplify the numerator and denominator factors:

- Numerator factor: $\sqrt{3\sqrt{17} \cdot 2}$. Wait, the term is $\sqrt{|x_{15}||x_{25}||x_{35}|} = \sqrt{\sqrt{17} \cdot 3 \cdot 2} = (6\sqrt{17})^{1/2} = \sqrt{6} \cdot 17^{1/4}$.
- Denominator factor 1: $\sqrt{|x_{14}||x_{24}||x_{34}|} = \sqrt{\sqrt{10} \cdot 2 \cdot 1} = (2\sqrt{10})^{1/2} = \sqrt{2} \cdot 10^{1/4}$.
- Denominator factor 2: $|x_{12}x_{23}x_{31}| = \sqrt{2} \cdot 1 \cdot \sqrt{5} = \sqrt{10}$.

Step 4.3: Final Expression

$$\mathcal{G}_1 = \frac{1}{2} \frac{\sqrt{6} \cdot 17^{1/4}}{(\sqrt{2} \cdot 10^{1/4})(\sqrt{10})}$$

$$\mathcal{G}_1 = \frac{1}{2} \frac{\sqrt{6} \cdot 17^{1/4}}{\sqrt{2} \cdot 10^{3/4}}$$

$$\mathcal{G}_1 = \frac{1}{2} \left(\frac{3}{10^{3/2}} \right)^{1/2} \frac{17^{1/4}}{\sqrt{2}} = \frac{1}{2} \sqrt{\frac{3}{10\sqrt{10}}} \frac{17^{1/4}}{\sqrt{2}}$$

Numerical approximation:
$\sqrt{6} \approx 2.449$
$17^{1/4} \approx 2.03$
$\sqrt{2} \approx 1.414$
$10^{3/4} \approx 5.62$

$\mathcal{G}_1 \approx 0.5 \times \frac{2.449 \times 2.03}{1.414 \times 5.62} \approx 0.5 \times \frac{4.97}{7.95} \approx 0.312$

Step 5: Computation for Case (2)

Coordinates: $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$.

Step 5.1: Calculate Distances

- $|x_{ij}| = |x_i - x_j|$. Since all points are real, $|x_{ij}| = |i - j|$.
- Cyclic product: $|x_{12}| = 1, |x_{23}| = 1, |x_{31}| = 2$. Product $P_{cyc} = 2$.
- Terms involving x_4, x_5:
  - $|x_{14}| = 3, |x_{15}| = 4$
  - $|x_{24}| = 2, |x_{25}| = 3$
  - $|x_{34}| = 1, |x_{35}| = 2$
  - $|x_{45}| = 1$

Step 5.2: Substitute into Model

$$\mathcal{G}_2 = \frac{1}{2} \frac{1}{1^{1/4}} \frac{\sqrt{(4)(3)(2)}}{\sqrt{(3)(2)(1)} \cdot (1)(1)(2)}$$

Note: Check signs. The formula uses moduli, so all are positive.

$$\mathcal{G}_2 = \frac{1}{2} \frac{\sqrt{24}}{\sqrt{6} \cdot 2}$$

Step 5.3: Simplify

$$\mathcal{G}_2 = \frac{1}{2} \frac{\sqrt{4 \cdot 6}}{2\sqrt{6}}$$

$$\mathcal{G}_2 = \frac{1}{2} \frac{2\sqrt{6}}{2\sqrt{6}}$$

$$\mathcal{G}_2 = \frac{1}{2}$$

Conclusion

The 5-point correlation function is modeled using the fermionic representation of the Ising CFT, resulting in an algebraic expression involving the metric distances between the points.

(1) Result for $x_1 = 1 + i, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$**:**

$$\langle \epsilon\epsilon\epsilon\sigma\sigma \rangle = \frac{\sqrt{3} \cdot 17^{1/4}}{2\sqrt{2} \cdot 10^{3/4}} \approx 0.312$$

(2) Result for $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$**:**

$$\langle \epsilon\epsilon\epsilon\sigma\sigma \rangle = \frac{1}{2}$$